

Sprint Week 1 – Winter 2026

Project Kick Off: February 20th

Project Due Date: March 6th

Welcome to Sprint Week everyone! In this sprint, you will be tasked with creating **two** separate Java programs. The **first** will be a server program that implements an API that is accessible via HTTP, and the **second** will be a client which connects via http to the server and will allow you to answer 4 questions making use of the APIs routes you build out

As always you are encouraged to pitch and implement your own projects, though for clarity here are some general requirements to follow for the sprint:

API (Server):

- Use **Spring Boot** to expose a **REST API** over **HTTP**.
- Store data in a **relational DB** (e.g. **MySQL**).
- Design the API so it can be fully tested in **Postman with manual testing**
- Should have the four relationships as outlined below
- Should have all HTTP verbs utilized in the API (GET, PUT, POST, DELETE)
- Project should be on GitHub with good evidence of branching and PR workflow on each repository
- The API should be in its own repo, separate from the CLI
- A project board should be set up and utilized to encourage team collaboration. For example: GitHub projects can be used

Client (CLI APP)

- Build a separate **Java CLI** program that connects to the API **via HTTP**.
- Only 4 GET requests are needed to answer the 4 specific questions below, it is not required for you to do any POST, PUT, PATCH OR DELETE from the CLI/Front End,
- Must be in its own repo separate from the API
- Set up GitHub actions on the front-end repo to run test when you make PRs and merge into your main
- Use JUnit for testing the front end, you should have mocking included in your test, we are looking for good test coverage so make sure to write lots!

NOTE: there is no need for a large dataset, 10s of records is fine

Demo Video:

- Create a video that demos your project, each team member should participate in talking about the project

- Showcase the features, talk about the good, talk about what went bad as a team

Example Project Data Model

There are 4 core entities:

- City: (id,name,state,population)
- Passenger: (id,firstName,lastName,phoneNumber)
- Airport:(id,name,code)
- Aircraft:(id, type, airlineName,numberOfPassengers)

Relationships

- A **City** has **many Airports**
- A **Passenger** lives in **one City** and can fly on **many Aircraft**
- An **Aircraft** can carry **many Passengers** and use **many Airports** (for takeoff/landing)
- An **Airport** belongs to **one City**

Questions Your App Must Answer (via CLI)

1. What airports are there in each city?
2. What aircraft has each passenger flown on?
3. What airports do aircraft take off from and land at?
4. What airports have passengers used?

Requirements for submission portal:

- GitHub links for both front end and back-end repos
- Link to your video for your team
- A small document outlining the sprint from your perspective, how it went, what was good, what was bad, what could you improve

Extra Resources:

- <https://github.com/JKells99?tab=repositories>
- <https://github.com/jamie-keyin>
- <https://docs.spring.io/spring-boot/index.html>
- CLI: [Mock Client Example](#) from class

Pass Outstanding Criteria

- Project goes above and beyond in some way
- There are 2 separate repositories one for CLI , one for API
- Good test coverage on CLI with a good amount of test written
- GitHub actions set up and passing on Front End
- Really good evidence of Team collaboration (Regular commits from all members, Branching/ PR workflow, project board set up)
- All 4 Questions are answered from your CLI application

